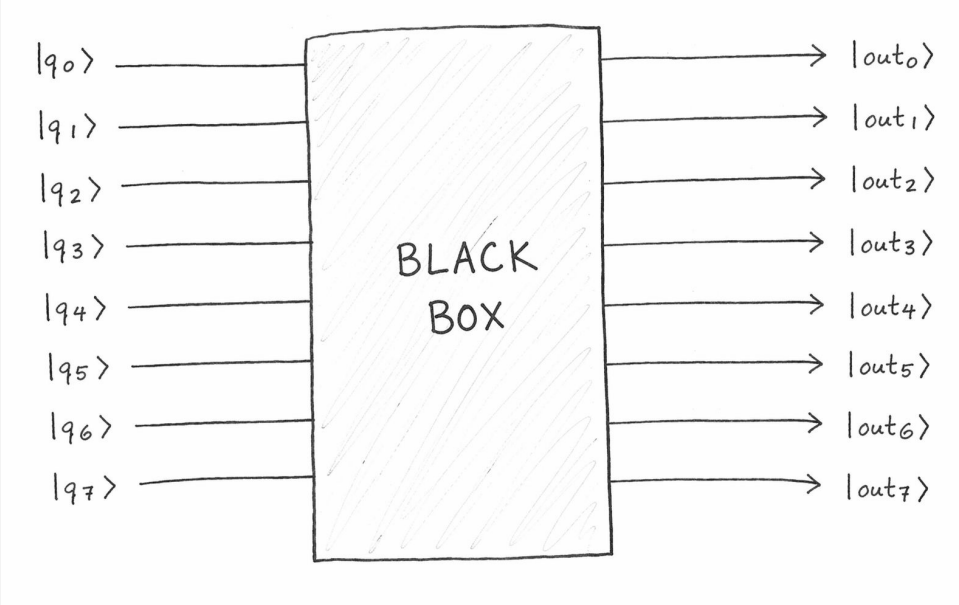
**YQuantum**

*HackYeah*

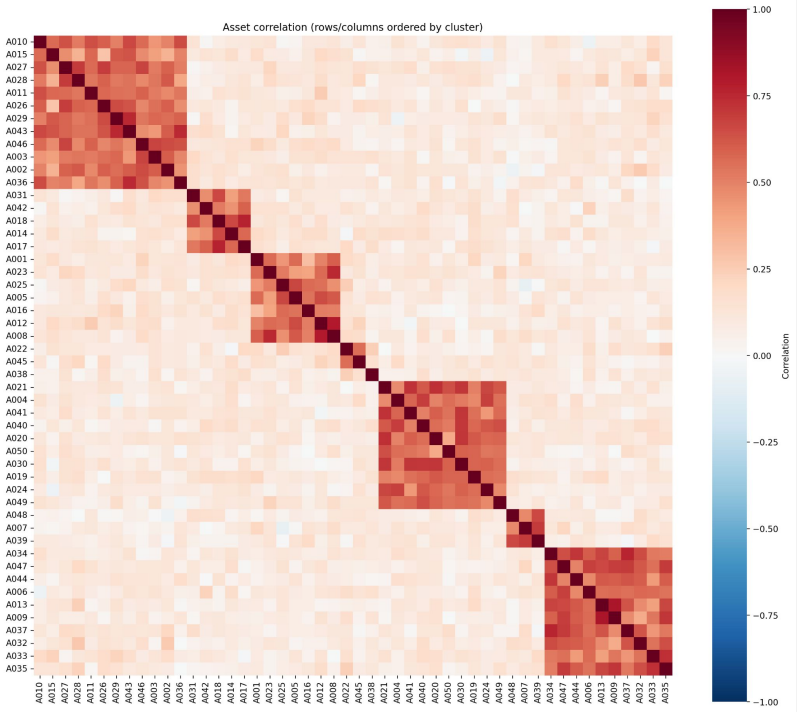
# Thought Process



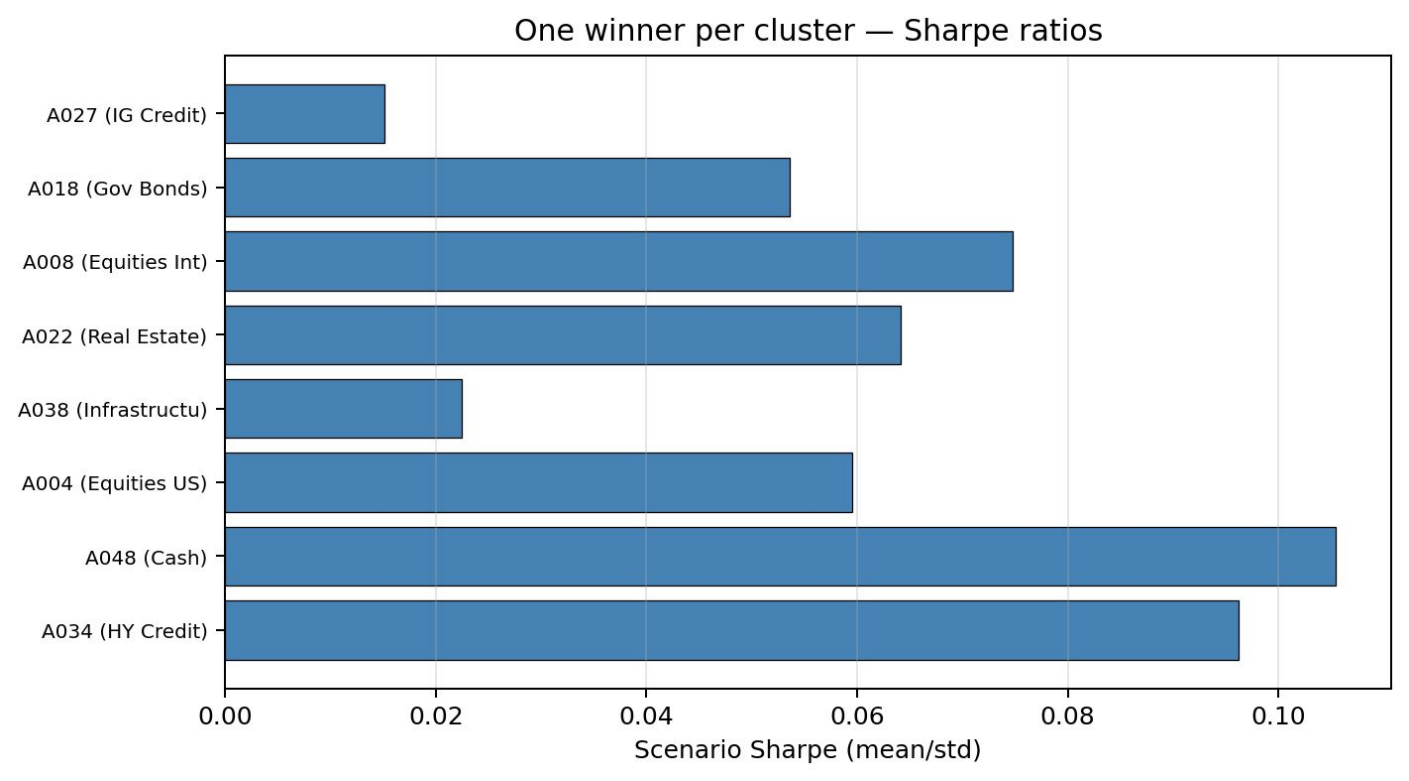

- Allowed 8 qubit gates
- Each qubit : Carries 1 bit of information when measured: Either buy or don't buy

# Cluster matrix

- Plot covariance matrix => Visualize clusters
- Find the “best” asset from each cluster
  - Maximize Sharpe Ratio (return/risk)



# Choosing “winners”

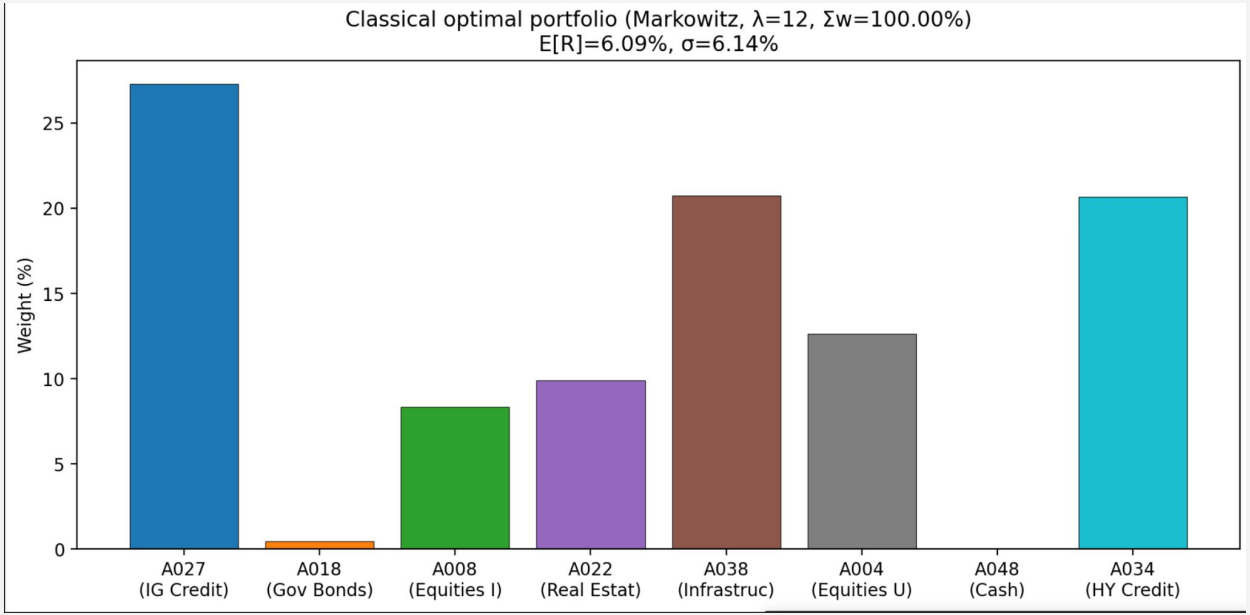


# Markowitz Weighting

$$\min q \sum_i \sum_j w_i w_j \sigma_{ij} - \sum_i w_i \mu_i$$

$$s.t. \sum_i w_i = 1$$

- Classical baseline for portfolio allocation
- Balances return and risk via λ
- Requires manual tuning of parameters



# From Portfolio Optimization to Physics

*From Portfolio Optimization to Physics*

- Goal: maximize return, minimize risk
- Classical → Markowitz optimization
- Quantum → energy minimization (ground state)
- Key idea: finance → physics mapping

*QUBO → Ising → Rydberg blockade→ Hamiltonian*

# QUBO Formulation

- Binary decision: xi∈0,1
- Diagonal → returns (μ)
- Off-diagonal → covariance (Σ)
- Constraint via penalty term

$$\max_{x} \quad \mu^T x - \lambda_{\text{risk}}\, x^T \Sigma x$$

$$\text{s.t.} \quad \sum_i x_i = B, \quad x_i \in \{0,1\}$$

$$E(x) = -\mu^T x + \lambda_{\text{risk}}\, x^T \Sigma x + \lambda_{\text{budget}} \left( \sum_i x_i - B \right)^2$$

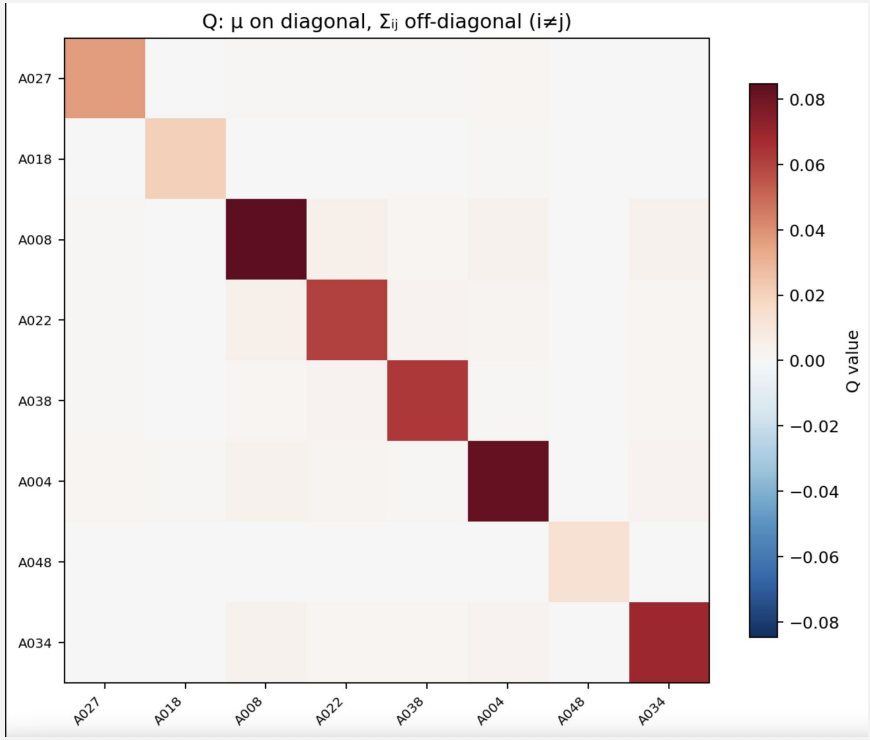
# The λ Problem

- Large λ → constraint satisfied, poor return
- Small λ → high return, invalid portfolio
- Requires manual tuning
- Not robust in practice

Click!

$$E(x) = -\mu^T x + \lambda x^T \Sigma x$$

# QUBO → Ising + Structure




$$x_i = \frac{1 + s_i}{2}, \quad s_i \in \{-1, +1\}$$

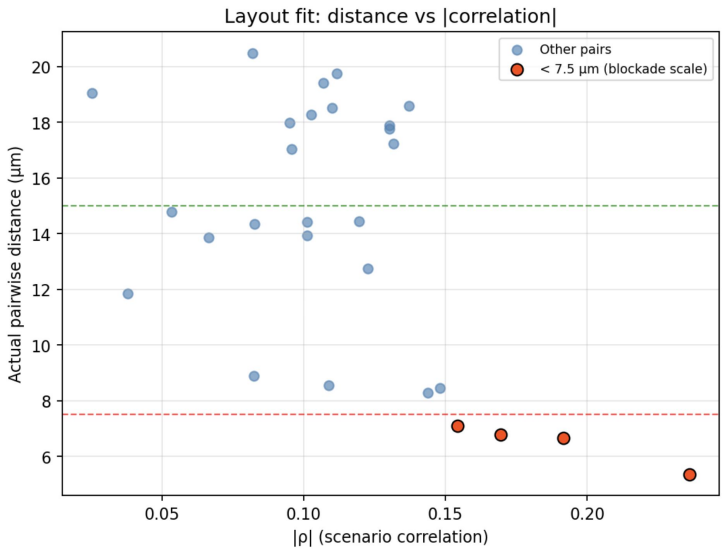
$$E(s) = C_0 + \sum_i h_i \, s_i + \sum_{i<j} J_{ij} \, s_i s_j$$

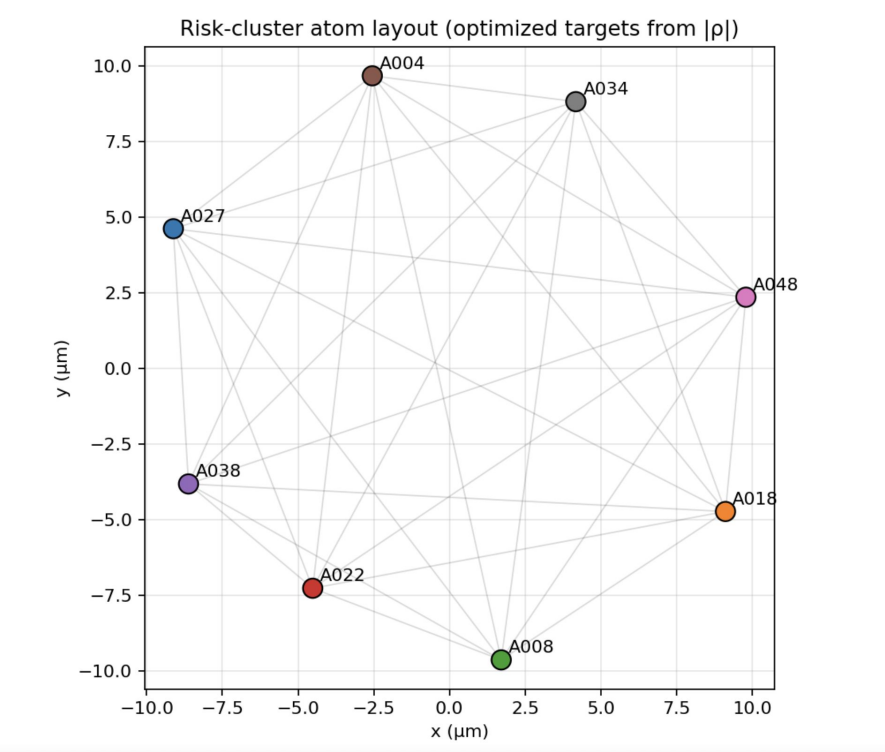
- QUBO encodes returns and pairwise interactions
- Diagonal → single-asset contribution
- Off-diagonal → covariance / coupling
- Converted to Ising form for quantum hardware

# Mapping to Rydberg System

- Correlation mapped to distance
- High correlation → atoms close
- Rydberg blockade prevents co-selection
- Constraint enforced physically (no λ)

$$V_{ij} = \frac{C_6}{R_{ij}^6}$$




https://www.quera.com/glossary/rydberg-blockage

# Quantum Dynamics: How Solution Emerges Click

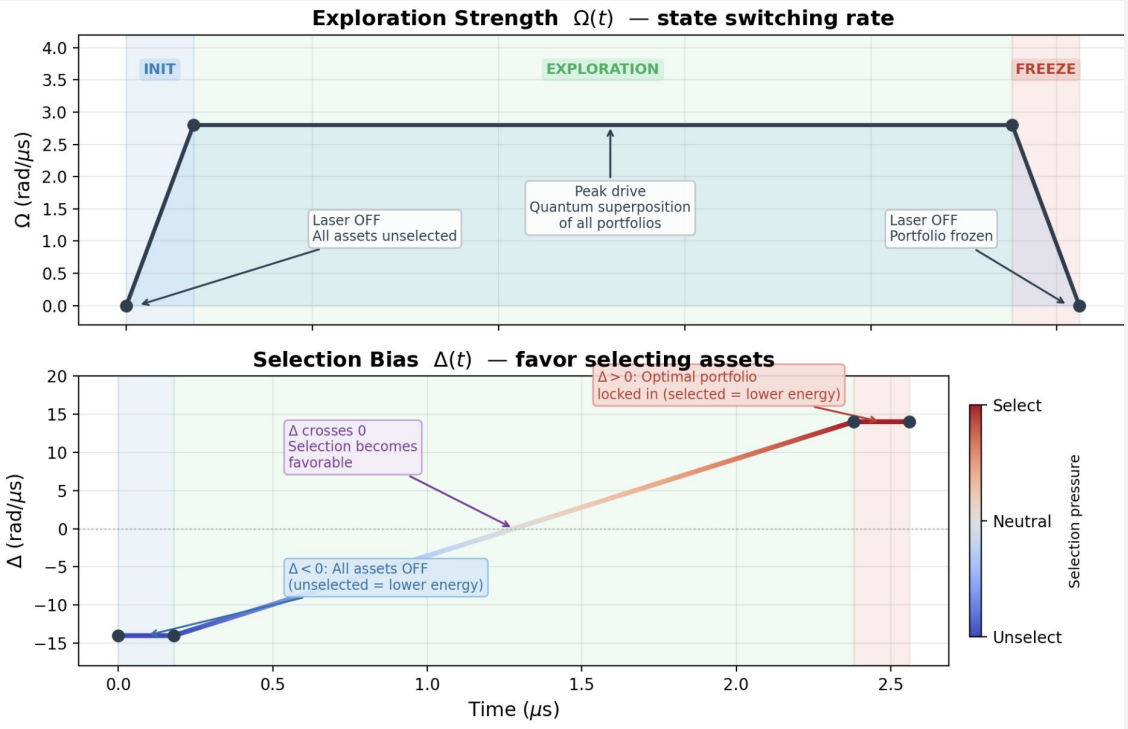
- Rabi ($\Omega$) → exploration (state transitions)
- Detuning ($\Delta$) → selection bias
- Interaction → correlation penalty
- Adiabatic evolution → ground state

$$H(t) = \frac{\Omega(t)}{2}\sum_i \sigma_x^{(i)} - \Delta(t)\sum_i n_i + \sum_{i<j} V_{ij} n_i n_j$$

$$E(x) = x^T Q x$$

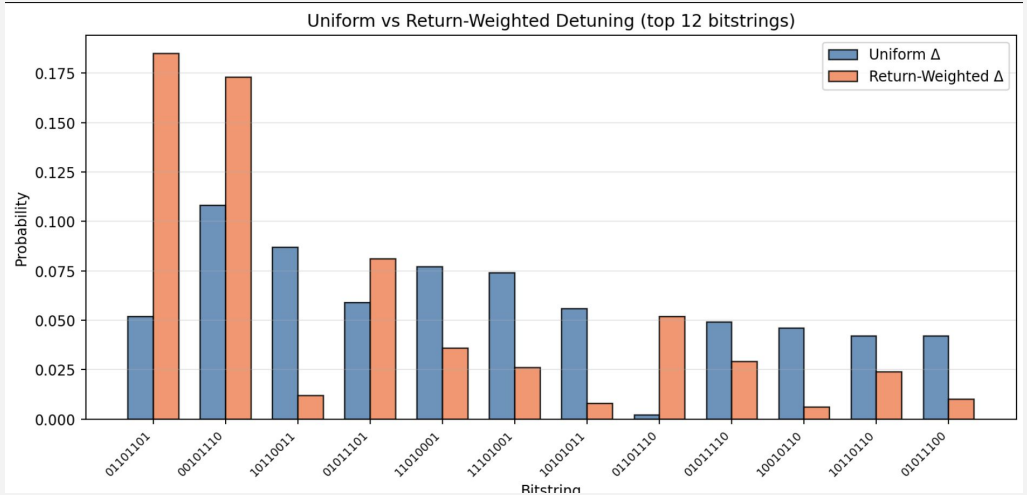
$$x_i = \frac{1+s_i}{2}$$

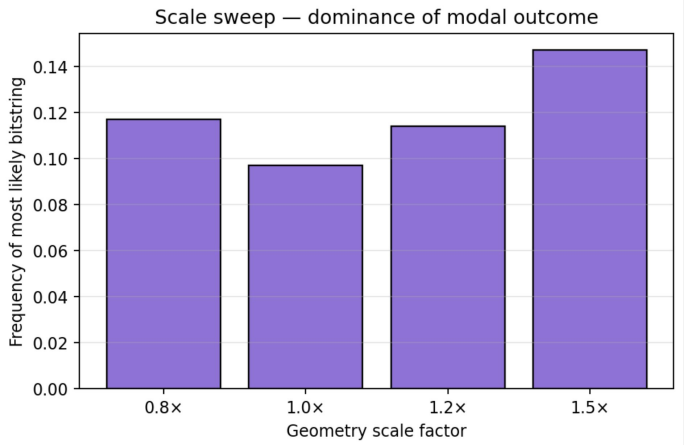
$$E(s) = C + \sum h_i s_i + \sum J_{ij} s_i s_j$$



# Results & Control via Physics

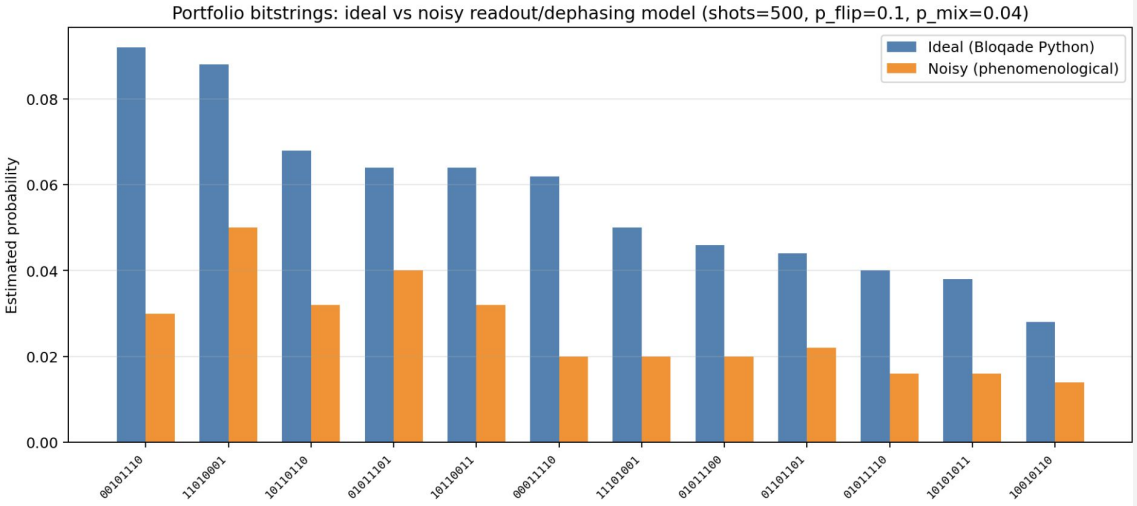
- Measurement → bitstring = portfolio
- Most frequent = lowest energy
- Geometry controls diversification
- Local detuning → return-aware selection



# Noise Resilience: Does the Quantum Signal Survive?

- Real hardware is noisy
- 10% bit-flip + 4% decoherence mix
- Noise increases entropy
- Top bitstrings still survive
- Hybrid signal remains robust



# Result

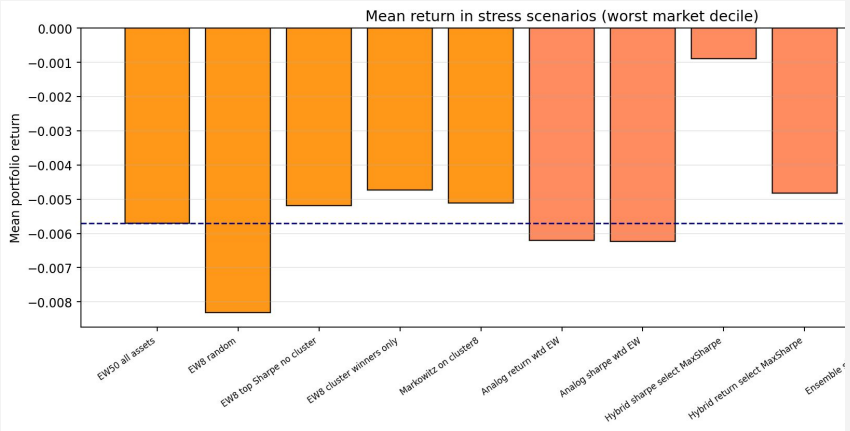
## Quantum-Enhanced Portfolio Optimization: Key Findings

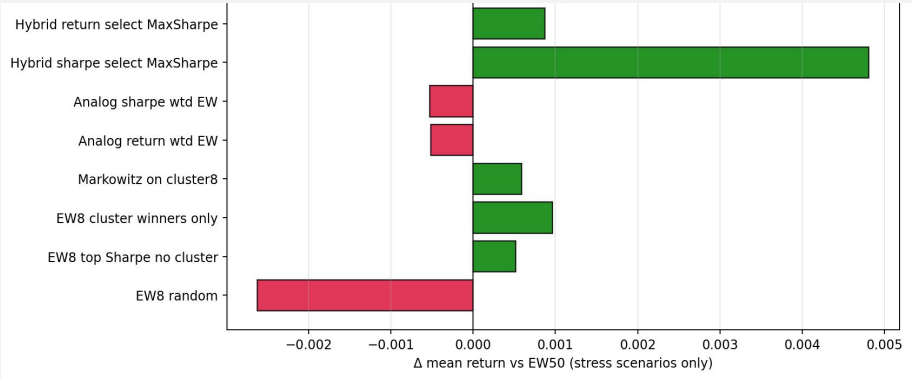
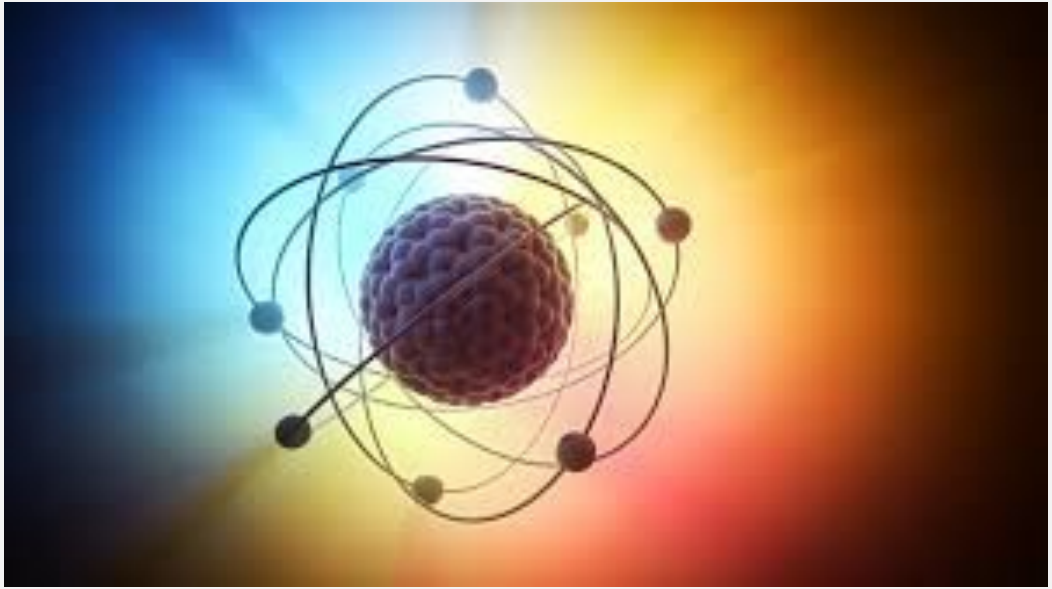
| Method | Sharpe (All) | CVaR 5% (All) | Sharpe (Stress) | Mean Return (All) |
|---|---|---|---|---|
| **EW50 (Baseline)** | 0.071 | -0.67% | -4.48 | 2.38 bps |
| **Classical Markowitz** | 0.107 | -0.78% | -1.83 | 4.17 bps |
| **Quantum Return-Weighted** | 0.114 | -0.93% | -1.98 | **5.30 bps** |
| **Quantum Sharpe-Weighted** | 0.094 | -0.98% | -1.80 | 4.64 bps |
| **Hybrid Quantum + Markowitz** | **0.131** | **-0.18%** | **-1.06** | 1.26 bps |
| **Ensemble Quantum** | 0.117 | -0.69% | -2.04 | 4.04 bps |

*Quantum Return-Weighted: +27% higher return than Classical Markowitz (5.30 vs 4.17 bps).*
*Hybrid Quantum + Markowitz: 73% lower tail risk, 3.4x better under stress.*
*Quantum selects which assets (physics); classical optimizes how much (math).*

# Economic Recession Case Study

We *simulated* economic stress (bottom 10%)






# Thank you!